

Lost In Translation: Uneven Grounding Degradation Under Cross-Lingual Shift in LLM-Assisted Cyber Threat Intelligence

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Abstract

We introduce **CrossCTI**, a multilingual benchmark for evaluating retrieval-augmented generation (RAG) grounding faithfulness in open-source large language models applied to cyber threat intelligence (CTI) analysis. Existing CTI and forensic NLP benchmarks are predominantly English-centric, leaving a critical gap in understanding how grounding faithfulness degrades under cross-lingual distribution shift, a realistic condition in multinational security operations centers. CrossCTI translates 50 English CTI reports into Arabic, Chinese, Swahili, and Russian using NLLB-200 with entity-preserving masking, achieving 100% IOC preservation across languages. We construct 26 forensic question-answer pairs and evaluate each against all five language corpora and three open-source models (Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B-v0.3), producing 390 evaluation instances, scored with token-level F1, character-level F1, and entailment-based attribution. Using a repeated-measures mixed-effects analysis that accounts for the fact that each of the 26 items is evaluated under all 15 model $\times$ language conditions, we find that cross-lingual degradation is severe, but *uneven across metrics*: language is the dominant, robust predictor of lexical (token-level) grounding quality ($p < 0.001$), with no detectable effect of model identity on this metric. In contrast, model identity is a significant predictor of both character-level overlap and entailment-based attribution ($p < 0.01$), including a specific interaction in one model (Mistral) that underperforms specifically in Swahili. Token-level F1 drops from 0.44 (English) to 0.03 (Arabic); gold-chunk retrieval recall drops from 0.962 (English) to 0.115 (Arabic); and the instance-level correlation between retrieval success and token-level grounding is $r = 0.60$. Performance also improves from the top- $k = 1$ to the top- $k = 3$ retrieved passages before plateauing, with no consistent gains from the additional retrieved content. We further show

that the entailment-based attribution metric is significantly elevated specifically for the model used to generate the benchmark’s own QA pairs, a synthetic-annotator confound that we quantify directly via within-item paired tests rather than only disclosing qualitatively. CrossCTI provides a reproducible framework for multilingual CTI grounding evaluation and a case study in how the apparent dominant factor, language or model, depends on which grounding metric is used.

1 Introduction

Retrieval-augmented generation (RAG) has emerged as a dominant paradigm for improving factual grounding in large language models (LLMs) by conditioning generation on externally retrieved evidence (Lewis et al., 2021). Despite these advances, RAG systems remain susceptible to hallucination and attribution failures, particularly under conditions of imperfect or insufficient retrieval (Shuster et al., 2021; Tang et al., 2024). Faithfulness, defined as the consistency between generated outputs and their retrieved evidence sources, has consequently become a central axis of RAG evaluation (Es et al., 2025; Tang et al., 2024). Existing evaluation frameworks, however, are predominantly developed and validated in English-centric, general-domain settings. This limits their ability to characterize grounding behavior under cross-lingual distribution shift, despite well-documented evidence that multilingual conditions introduce substantial degradation in both retrieval and generation quality (Asai et al., 2021; Zhang et al., 2023). Critically, existing multilingual benchmarks primarily target either retrieval effectiveness (Zhang et al., 2023) or cross-lingual question answering (Asai et al., 2021), without explicitly measuring the faithfulness of generated outputs with respect to retrieved evidence. More recent multilingual RAG evaluation efforts, such as MEMERAG (Blandón et al., 2025), extend

evaluation to multilingual settings using human judgments, but remain anchored to general-domain Wikipedia corpora and do not address structured, domain-specific reasoning under retrieval grounding constraints. This gap is especially consequential in cyber threat intelligence (CTI), where RAG systems are increasingly explored for automating analysis of threat reports, extracting indicators of compromise (IOCs), and supporting digital forensics and incident response (DFIR) workflows (Clairoux-Trepanier et al., 2024; Alhuzali, 2025). Recent work has shown that LLMs exhibit unreliable behavior on real-scale CTI reports, demonstrating inconsistency and overconfidence even in monolingual English settings (Mezzi et al., 2025). Hallucinated or misattributed threat intelligence poses a direct operational risk: an erroneous IOC attribution or a fabricated TTP mapping can misdirect incident response at critical decision points. These failure modes are substantially amplified under cross-lingual conditions, yet no benchmark has characterized their extent.

To address this gap, we introduce **CrossCTI**, a multilingual benchmark for evaluating RAG grounding faithfulness under cross-lingual distribution shift in CTI analysis. We construct five language corpora by retaining 50 English CTI reports and translating them into Arabic, Chinese, Swahili, and Russian using NLLB-200 (Team et al., 2022), with entity-preserving masking applied to maintain IOC integrity across all language pairs. We construct 26 forensic QA pairs and evaluate each across all five language corpora and three open-source instruction-tuned models, Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B-v0.3, providing 390 evaluation instances (26 items $\times$ 5 languages $\times$ 3 models). Because every item is repeated across all 15 model $\times$ language conditions, we treat this explicitly as a repeated-measures design and analyze it with item-level mixed-effects models, cluster-robust regression, and within-item nonparametric tests, rather than treating the 390 instances as independent observations.

Our central finding is that *which factor appears to dominate depends on the grounding metric*. Under lexical (token-level) F1, language is overwhelmingly dominant and model identity has no detectable effect, a result that holds consistently across a mixed-effects model, a cluster-robust regression, and a nonparametric Friedman test. Under character-level F1 and entailment-

based attribution, however, model identity is a significant factor in its own right, including a specific model $\times$ language interaction (Mistral underperforming in Swahili), and a significant, consistent elevation of the model used to generate the benchmark’s QA pairs on the attribution metric, a confound we quantify rather than only acknowledge. CrossCTI provides a reproducible framework for evaluating these failure modes and a case study in how metric choice shapes which axis of a RAG system, the language pipeline or the generator model, looks responsible for grounding failure.

2 Related Work

Retrieval-Augmented Generation and Faithfulness. RAG (Lewis et al., 2021) conditions generation on retrieval evidence to reduce hallucination (Shuster et al., 2021), but faithfulness evaluation remains an open problem. Frameworks such as RAGAs (Es et al., 2025) and MiniCheck (Tang et al., 2024) provide automated faithfulness scoring for English text; we adopt the general entailment-based faithfulness paradigm these frameworks establish but instantiate it with a multilingual NLI model (Section 3.3) rather than an English-only scorer, given our cross-lingual setting. Salemi and Zamani (2024) show that retrieval quality is itself a strong predictor of downstream RAG performance. In a security-specific setting, Fayyazi et al. (2025) introduce an embedding-based attribution method to quantify how much LLM responses in vulnerability analysis actually rely on retrieved evidence versus parametric knowledge, a complementary direction to our entailment-based faithfulness scoring.

Multilingual and Cross-Lingual RAG. MIR-ACL (Zhang et al., 2023) and XOR-TyDi QA (Asai et al., 2021) established that retrieval effectiveness and question answering accuracy both degrade under cross-lingual conditions, particularly for low-resource languages. More recently, XRAG (Liu et al., 2025) identifies that cross-lingual RAG systems struggle with both response language correctness under monolingual retrieval and with reasoning over multilingual retrieved evidence. At the same time, Li et al. (2025) show that incorporating multilingual document perspectives can improve robustness on culturally sensitive tasks, though citation behavior varies substantially by language resource level. MEMERAG (Blandón et al., 2025) extends multilingual RAG evaluation using human faithfulness judgments across five lan-

guages. CrossCTI differs from these efforts in two respects: it targets a structured, domain-specific reasoning task rather than open-domain Wikipedia or news QA, and it isolates language as an experimental variable independent of model choice via controlled translation rather than relying on naturally occurring multilingual corpora.

LLMs for Cyber Threat Intelligence. LLMs are increasingly applied to CTI workflows, including threat report summarization and IOC extraction (Clairoux-Trepanier et al., 2024). However, Mezzi et al. (2025) demonstrate that LLMs exhibit inconsistency and overconfidence on real-scale CTI reports even in monolingual English settings. No prior work has measured how these reliability features compound under cross-lingual distribution shift, which is the gap CrossCTI addresses.

3 CrossCTI Benchmark

3.1 Dataset and Translation Pipeline

CrossCTI is built from the mrminhaz/CTI-Reports dataset, a collection of publicly available English-language cyber threat intelligence reports sourced from open threat intelligence feeds. We load 100 reports and randomly sample 50 for evaluation using a fixed seed (seed = 42), reserving the remainder for corpus construction. Each report is segmented into overlapping chunks of 200 words with a 30-word stride, yielding a retrieval corpus of fixed-context passages that preserve local semantic coherence across chunk boundaries. To construct multilingual variants, we translate all English chunks into Arabic (arb_Arab), Chinese (zho_Hans), Swahili (swh_Latn), and Russian (rus_Cyr1) using NLLB-200 (Team et al., 2022), a massively multilingual neural machine translation model supporting 202 languages. Languages were selected to span a range of typological diversity, script systems, and resource levels: Arabic introduces right-to-left script and is high-resource but underrepresented in CTI corpora; Chinese provides a logographic script with strong coverage in Qwen2.5; Swahili represents a low-resource African language with limited forensic NLP precedent; and Russian provides a high-resource Cyrillic baseline with a natively documented threat actor ecosystem.

A critical preprocessing step is *entity-preserving masking*: prior to translation, all indicators of compromise (IOCs) — including IPv4 addresses, MD5, and SHA hashes, CVE identifiers, and file system

paths — are replaced with indexed placeholders (`__IOCPH{i}__`) using a regular expression pattern, translated text is generated, and placeholders are restored in post-processing. This prevents NLLB-200 from corrupting domain-critical tokens that carry no translatable semantic content. Entity preservation is verified post-hoc by checking whether each original IOC string appears verbatim in the translated output. CrossCTI achieves a 100% IOC preservation rate across all four target languages.

3.2 QA Pair Construction

We construct question-answer pairs using Qwen2.5-7B-Instruct as a synthetic annotator. The model is prompted under a structured system instruction to generate forensic QA pairs grounded strictly in sampled English chunk text, prioritizing questions about malware families, CVEs, IOCs, MITRE ATT&CK TTPs, targeted sectors, persistence mechanisms, and command-and-control infrastructure. Each pair consists of a question, a gold answer, and a verbatim source quote supporting the answer.

After generation and filtering, this process yields **26 valid QA pairs**. Each pair is evaluated against all five language corpora (English plus four translations) and all three candidate models, yielding $26 \times 5 \times 3 = 390$ total evaluation instances, or 26 instances per model $\times$ language cell. Because the same 26 items recur across every cell, instances are not statistically independent of one another; we treat this explicitly as a repeated-measures design throughout Section 4 rather than assuming 390 independent observations. A subset of 20 of the 26 pairs was manually reviewed for quality (Section 6).

We acknowledge that using Qwen2.5-7B-Instruct as both the QA generator and as one of the evaluated models constitutes a potential confounder. Section 4.3 presents direct statistical evidence of its effect on the attribution metric rather than treating it only as a disclosed risk.

3.3 Evaluation Protocol

Retrieval. For each language, we build a separate FAISS flat inner-product index (Johnson et al., 2017) over chunk embeddings produced by all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), with embeddings L2-normalized before indexing. At inference time, the top- k chunks most similar to the query are concatenated into a structured prompt. We evaluate at $k \in \{1, 2, 3, 5, 10\}$; primary results are reported at $k = 3$. All three

models share this single retriever; we return to this as a limitation in Section 6.

Generation. Three open-source instruction-tuned models are evaluated: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3. All models are loaded in 4-bit NF4 quantization via BitsAndBytes and prompted identically with a fixed zero-shot RAG template instructing the model to answer using only the retrieved sources. Generation uses greedy decoding ($T = 0$, $\text{max_new_tokens} = 128$) across all models to ensure comparability. We did not evaluate at full precision; this is discussed as a limitation.

Grounding Metrics. We report three complementary metrics. *Token-level F1* measures lexical overlap between the model answer and the gold answer at the unigram level. *Character-level F1* measures overlap at the character n-gram level, which is more robust to tokenization and morphological differences across scripts (Section 4.5). *Attribution* is assessed using XLM-RoBERTa-large fine-tuned on XNLI (Conneau et al., 2018, 2020) (the joeddav/xlm-roberta-large-xnli checkpoint), a multilingual natural language inference model natively covering all five evaluated languages as part of its 15-language XNLI fine-tuning set. The model outputs entailment, neutral, and contradiction probabilities for a (context, answer) pair; a generation is classified as hallucinated when its entailment probability falls below 0.5. We chose a multilingual NLI model rather than an English-only faithfulness scorer specifically to avoid introducing an additional cross-lingual validity gap on top of the retrieval-encoder gap already present in our pipeline (Section 6). Retrieval recall is computed as the fraction of instances in which the gold source chunk appears among the top- k retrieved passages.

4 Results

4.1 Overall and Per-Language Performance

Table 1 reports mean performance by language across all three metrics. All three metrics show a large English-vs-non-English gap, but the *shape* of degradation across non-English languages differs by the metric: Arabic is weakest on token F1, but Arabic and Chinese are statistically indistinguishable and jointly weakest on character F1 (Section 4.2), while Swahili shows a distinctly smaller character-level gap than its token-level gap would

Language	Token F1	Char. F1	Attrib.
English	0.441	0.683	0.590
Arabic	0.032	0.134	0.295
Russian	0.047	0.166	0.282
Swahili	0.079	0.468	0.333
Chinese	0.107	0.130	0.333

Table 1: Mean grounding scores by language, averaged across all three models ($n = 78$ per language). Arabic is the weakest language on token F1; Arabic and Chinese are statistically indistinguishable and jointly weakest on character F1.

Model	Token F1	Char. F1	Attrib.
Llama	0.131	0.327	0.308
Mistral	0.140	0.252	0.269
Qwen	0.154	0.369	0.523

Table 2: Mean grounding scores by model, averaged across all five languages ($n = 130$ per model).

suggest (Section 4.5).

Table 2 shows that model-level differences are small for token F1 but substantially larger for character F1 and, especially, attribution — where Qwen’s mean is roughly double Mistral’s. Section 4.3 shows this gap is not incidental.

4.2 Statistical Analysis: Language and Model Effects

Because each of the 26 QA items is evaluated under all 15 model×language conditions, observations sharing an item are not independent. We therefore report three converging analyses per metric: an item-level mixed-effects model with a random intercept for item, a fixed-effects regression with standard error clustered by item, and a nonparametric within-item Friedman test. All three approaches agree on the qualitative conclusion for each metric.

Token F1. Language is highly significant (e.g., English vs. Arabic coefficient 0.396, $p < 0.001$ in the mixed model), while model identity is not significant under any of the three approaches (mixed model $p = 0.53$ – 0.95 ; clustered regression $p = 0.18$ – 0.88 ; Friedman $\chi^2 = 2.38$, $p = 0.30$). No model×language interaction is significant. This is the one metric on which the original hypothesis, that language dominates and model identity does not matter, holds robustly.

Character F1. Language remains highly significant, but model identity is also now significant

Model	AR	EN	RU	SW	ZH
Llama	0.024	0.420	0.051	0.062	0.095
Mistral	0.027	0.438	0.048	0.075	0.112
Qwen	0.046	0.464	0.043	0.100	0.114

Table 3: Token-level F1 by model and language ($n = 26$ per cell).

(clustered regression: Qwen coefficient 0.0415, $p = 0.001$; Friedman $\chi^2 = 29.77$, $p < 0.001$), and there is a significant model $\times$ language interaction: Mistral underperforms specifically in Swahili relative to its own baseline pattern (interaction coefficient = -0.249 , $p < 0.001$, replicated in both the mixed model and the clustered regression). Within-item paired tests (Holm-corrected) confirm all three pairwise model comparisons are significant on this metric (all $p < 0.001$).

Attribution. Language remains significant (Friedman $\chi^2 = 11.21$, $p = 0.024$), and model identity is also significant when comparisons are pooled across five languages (Friedman $\chi^2 = 20.89$, $p < 0.001$), driven specifically by Qwen scoring higher than both other models (within-item paired tests, Holm-corrected Qwen vs. Llama and Qwen vs. Mistral both $p < 0.001$; Llama vs. Mistral not significant). We discuss the source of this effect in Section 4.3.

4.3 The Synthetic-Annotator Confound in Attribution

Qwen2.5-7B-Instruct is both the QA-pair generator (Section 3.2) and one of the three evaluated models. Table 2 shows Qwen’s mean attribution score (0.523) is roughly double Mistral’s (0.269) and well above Llama’s (0.308), while all three models are statistically indistinguishable on token F1. This pattern, an advantage specific to the metric closest to “does this answer match what I would have generated,” and specific to the model that generated the reference material, is the expected signature of QA-generator confound rather than a genuine capability difference. Within-item paired testing makes this precise: Qwen’s advantage over each other model on attribution is significant after Holm correction ($p < 0.001$ in both comparisons), while Llama and Mistral are never distinguishable from each other on any of the three metrics. We do not use the attribution metric to support any claim about relative model capability, and recommend that future extensions of CrossCTI generate

Language	Retrieval Recall
English	0.962
Arabic	0.115
Russian	0.269
Swahili	0.308
Chinese	0.192

Table 4: Gold-chunk retrieval recall by language ($n = 26$ per language; identical across all three models, as all share a single retriever).

Language	Token F1	Char. F1	Ratio
Swahili	0.079	0.468	$5.92\times$
Arabic	0.032	0.134	$4.11\times$
Russian	0.047	0.166	$3.51\times$
English	0.441	0.683	$1.55\times$
Chinese	0.107	0.130	$1.21\times$

Table 5: Ratio of character-level to token-level F1 by language. A high ratio indicates token-level F1 substantially underestimates a true overlap relative to character-level overlap.

QA pairs with a model held out from evaluation entirely.

4.4 Instance-Level Retrieval-Grounding Relationship

Table 4 shows retrieval recall collapses outside English, most severely for Arabic. At the instance level ($n = 390$), retrieval success is significantly correlated with all three grounding metrics: token F1 ($r = 0.60$, $p < 10^{-39}$), character F1 ($r = 0.58$, $p < 10^{-35}$), and attribution ($r = 0.31$, $p < 10^{-9}$). Mean token F1 is 0.047 when the gold chunk is not retrieved versus 0.303 when it is, a more than sixfold difference.

4.5 Cross-Lingual Tokenization Sensitivity

Table 5 shows that the gap between token-level and character-level F1 is far from uniform across languages. For Swahili, an agglutinative Bantu language, character F1 is nearly six times token F1, indicating that whitespace/morpheme-boundary tokenization substantially undercounts genuine lexical overlap. Arabic and Russian, both morphologically rich, show similar if smaller effects. Chinese and English, by contrast, show token F1 and character F1 in much closer agreement. This indicates that a meaningful share of Swahili’s apparent token-level “failure” in Table 1 is a tokenization artifact rather than a grounding failure, while Chinese and Arabic’s low scores are consistent across both granularities and more likely reflect genuine

grounding degradation. We recommend that future multilingual RAG evaluations report both token- and character-level overlap, particularly when morphologically rich or agglutinative languages are included.

4.6 Top- k Sensitivity

Performance improves from top- $k = 1$ (0.199) to top- $k = 3$ (0.369), after which it plateaus (top- $k = 5$: 0.370; top- $k = 10$: 0.370), indicating diminishing returns from additional retrieved context beyond $k = 3$. This analysis was conducted at the level of the full 390-instance dataset.

5 Discussion

Our results reveal that retrieval quality is a strong, statistically robust predictor of grounding faithfulness at the instance level (Section 4.4), and its collapse under cross-lingual shift, driven by an English-centric encoder (all-MiniLM-L6-v2), plausibly explains a large share of the degradation observed on token-level F1. However, our repeated-measures analysis shows this is not the whole story: on character-level overlap and entailment-based attribution, model identity is an independent, statistically significant factor, including a specific and replicated Mistral×Swahili weakness that a retrieval-only account would not predict. This suggests that at least part of the multilingual grounding gap in CTI-focused RAG is attributable to generation behavior, not retrieval alone, and that conclusions about “language versus model” in cross-lingual RAG evaluation are sensitive to which grounding metric is used.

The tokenization-sensitivity analysis (Section 4.5) reinforces this point from a different angle: a meaningful share of the apparent Swahili grounding failure under token F1 is a measurement artifact of whitespace-based tokenization, not a genuine grounding failure, while Chinese and Arabic’s weak performance is consistent across both token- and character-level measurement and more likely reflects a real effect. We recommend character-level (or morphology-aware) metrics as a standard complement to token F1 in future multilingual RAG evaluation.

Cross-lingual validity of the attribution metric. Because we use a multilingual NLI model fine-tuned on XNLI rather than an English-only faithfulness scorer, we are able to check the attribution metric’s internal and external consistency separately

in each evaluated language, rather than only in English. The derived binary attribution label is internally consistent with the underlying continuous entailment probability in every language (point-biserial $r = 0.86$ – 0.94 , all $p < 10^{-23}$; Arabic $r = 0.896$, English $r = 0.941$, Russian $r = 0.907$, Swahili $r = 0.875$, Chinese $r = 0.859$), as expected since both are derived from the same model output; we treat this as a sanity check rather than independent validation. The more informative cross-lingual check is the model’s relationship to a fully independent signal, token-level F1, which does not draw on the NLI model at all. Its relationship to lexical overlap is weaker and more variable, and breaks down specifically for Chinese: Spearman correlation between entailment score and token F1 is significant for Arabic, English, Russian, and Swahili ($\rho = 0.28$ – 0.37 , $p < 0.012$) but not for Chinese ($\rho = 0.170$, $p = 0.136$). We read this as a real but narrow caveat rather than evidence against the metric generally: a multilingual NLI model can be internally consistent within a language while still disagreeing with a lexical-overlap metric for reasons specific to that language’s tokenization or script (Section 4.5), and this is a difference we can only observe here because the underlying model was designed for multilingual use in the first place.

Finally, the attribution metric’s elevation for Qwen, the model used to generate the benchmark’s own QA pairs, is a caution about self-referential benchmark construction rather than a finding about Qwen’s capability. We isolate it statistically here, so it cannot be mistaken for a genuine three-way model comparison; future extensions of CrossCTI should generate QA pairs with a model held out from the evaluated set entirely.

6 Limitations

Sample size and repeated measures. CrossCTI evaluates **26** unique QA items, each replicated across 5 languages and 3 models (390 instances total). Because the same 26 items recur across every model×language cell, the 390 instances are not independent observations, and we report mixed-effects, cluster-robust, and nonparametric within-item tests throughout Section 4 rather than treating instance-level tests as the primary inferential model. This repeated-measures structure limits statistical power for finer-grained analyses (e.g., per-question-type breakdowns) more than the raw instance count alone would suggest, and we recommend that fu-

ture extensions of CrossCTI substantially increase the number of unique underlying items.

Synthetic QA construction. QA pairs are generated by Qwen2.5-7B-Instruct, which is also one of three evaluated models. We no longer treat this as only a disclosed risk: Section 4.3 shows a statistically significant, Holm-corrected elevation of Qwen’s attribution relative to both other models, consistent with this confound. We manually reviewed 20 of the 26 pairs for quality, but this does not address the annotator-identity confound itself. We do not use the attribution metric to support cross-model capability claims, and recommend that future work generate QA pairs with a model held out from the evaluated set.

Single retriever and encoder; no oracle-retrieval condition. All experiments use a single embedding model (all-MiniLM-L6-v2) and a single dense retrieval strategy, shared identically across all three generator models. The observed language effect on retrieval recall (Table 4) may therefore be partially attributable to this encoder’s multilingual coverage rather than a universal property of cross-lingual RAG. We did not run a force-fed gold-context (“oracle retrieval”) generation condition that would isolate generation-layer faithfulness from retrieval-layer failure; the instance-level correlation in Section 4.4 is suggestive but not a substitute for this experiment, which we identify as the highest-priority extension to this work.

Additional scope limitations. Several further factors bound the generalizability of these results. All three generator models were evaluated exclusively under 4-bit NF4 quantization; we did not evaluate at full precision (FP16/BF16), and since aggressive quantization is known to disproportionately affect multilingual representations, we cannot rule out that some of the reported cross-lingual degradation, or some of the model-level effects reported in Section 4.2, would shift under full-precision inference. Relatedly, all three evaluated models are in the 7–8B parameter range, so our finding that model identity does not explain token-level grounding degradation should be read as scoped to this size range rather than generalized to substantially larger models. On the data side, while entity-preserving masking achieves 100% IOC preservation, we do not independently verify the semantic fidelity of NLLB-200 translations beyond entity-level checks, and translation artifacts

in domain-specific CTI terminology may introduce noise that is conflated with genuine cross-lingual grounding failure; more broadly, CrossCTI uses machine-translated English reports rather than natively authored multilingual CTI documents, which may affect the generalizability of our findings to real-world multilingual SOC environments.

7 Future Work

The highest-priority extension is a force-fed gold-context (oracle-retrieval) generation condition, which would directly separate retrieval-layer failure from generation-layer faithfulness rather than relying on the instance-level correlation reported here. We also plan to evaluate with a multilingual-specific retrieval encoder to test whether the observed language effect is encoder-specific; regenerate QA pairs using a model held out from evaluation to remove the annotator confound identified in Section 4.3; evaluate at full precision to rule out quantization-driven effects; scale to a larger, more diverse set of underlying QA items to improve statistical power under the repeated-measures design; and extend evaluation to native (non-translated) multilingual CTI corpora.

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